Wenbo Lyu

B.Sc. Graduate

Curriculum Vitae
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Education and qualifications

B.Sc. In Geographic Information Science   Shaanxi Normal University   2025

Research Experience

2026.6-	Research Assistant, The Hong Kong Polytechnic University
2025.9-2026.6	Research Assistant, Eastern Institute of Technology, Ningbo
2024.8-2025.8	Research Intern, The Hong Kong University of Science and Technology (Guangzhou)

Research interests

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.

Selected research papers

Lyu W, Dai S, Song Y, Zhao W, Yi W, Xiao Y, Jia N (2026). "Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality." *International Journal of Geographical Information Science*, 1-23. ISSN 1362-3087, doi:10.1080/13658816.2026.2687121 https://doi.org/10.1080/13658816.2026.2687121.

Lyu W, Lei Y, Yi W, Song Y, Li X, Dai S, Qin Y, Zhao W (2026). "Causal discovery in urban data with temporal empirical dynamic modeling: The R package tEDM." *Computers, Environment and Urban Systems*, *127*, 102435. ISSN 0198-9715, doi:10.1016/j.compenvurbsys.2026.102435 https://doi.org/10.1016/j.compenvurbsys.2026.102435.

Lv W, Lei Y, Liu F, Yan J, Song Y, Zhao W (2025). "gdverse: An R Package for Spatial Stratified Heterogeneity Family." *Transactions in GIS*, *29*(2), e70032. ISSN 1467-9671, doi:10.1111/tgis.70032 https://doi.org/10.1111/tgis.70032.

Lv W, Liu F, Cai K, Cao Y, Deng M, Liang W, Yan J, Wang G (2024). "Distinguishing the Impacts and Gradient Effects of Climate Change and Human Activities on Vegetation Cover in the Weihe River Basin, China." *Journal of Geophysical Research: Biogeosciences*, *129*(10), e2024JG008297. ISSN 2169-8961, doi:10.1029/2024jg008297 https://doi.org/10.1029/2024jg008297.

Li B, Lv W, Ding H, Song Y, Zhao W (2026). "New-type urbanization policy, urban expansion, and employment quality in chinese cities: evidence from a quasi-natural experiment in China." *GIScience & Remote Sensing*, *63*(1), 2716413. doi:10.1080/15481603.2026.2716413 https://doi.org/10.1080/15481603.2026.2716413.

Xu L, Shi H, Lyu W, Zhang Y, Zhao H (2026). "Beyond correlation: Unveiling the causal link between street-scapes and recreational walking/cycling." *Journal of Transport Geography*, *136*, 104789. ISSN 0966-6923, doi:10.1016/j.jtrangeo.2026.104789 https://doi.org/10.1016/j.jtrangeo.2026.104789.

Chen C, Song Y, Lv W, Shemery A, Hampson K, Yi W, Zhong Y, Wu P (2025). "Predicting pavement cracking performance using laser scanning and geocomplexity-enhanced machine learning." *Computer-Aided Civil and Infrastructure Engineering*. ISSN 1467-8667, doi:10.1111/mice.13489 https://doi.org/10.1111/mice.13489.

Note: Publications under "Lyu, W." and "Lv, W." both refer to Wenbo Lyu.